

Test Plan Document

By Team 74

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1.0 Project Information

1.1 Document Purpose

The purpose of this document is to outline a test plan for the 'View All Routes' page in the Domestic Travel Co. application. In particular, the document illustrates all the necessary system/integration tests as well as unit tests needed to make sure that the 'View All Routes' page is working properly and that the code/functions used are working as intended.

1.2 Specification of the Feature Being Tested

The feature being tested in the 'View All Routes' feature.

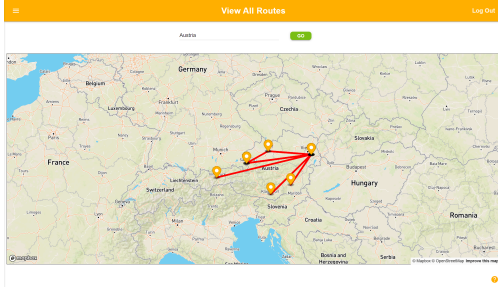
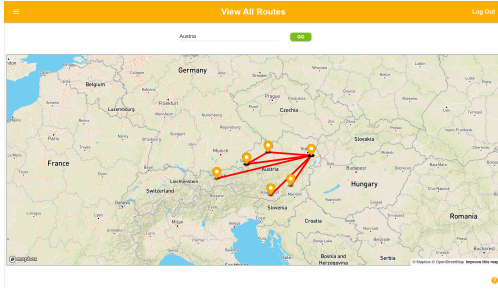
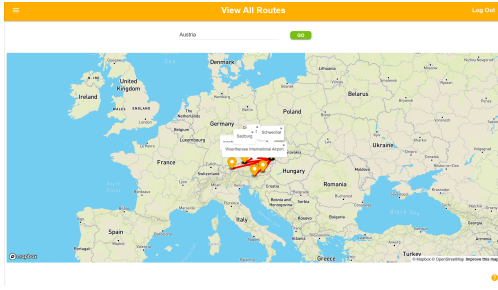
1.2.1 Description of Feature

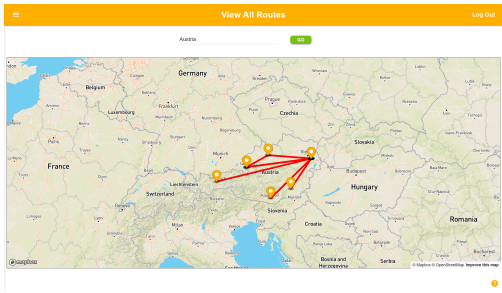
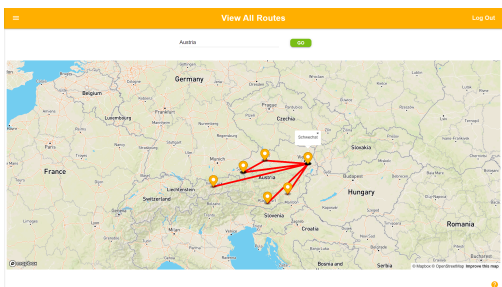
The 'View All Routes' feature allows the user to search for a country and view all the routes in that particular country. All the routes shown in this 'View All Routes' page have to be domestic routes. Additionally, the feature allows the user to get the airport's name by clicking on the marker, however, does not allow the user to get specific route details because the user cannot plan any trips on this page and instead, this page is supposed to only provide a visual of all the possible routes the user can go on within a country.

1.2.2 Acceptance Criteria

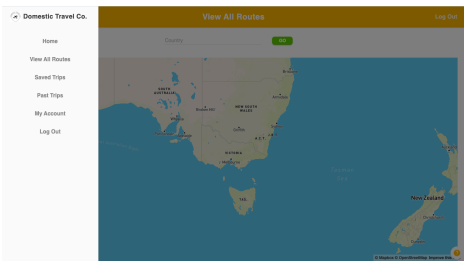
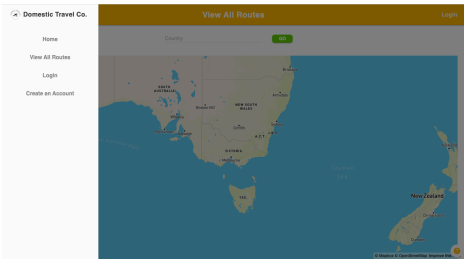
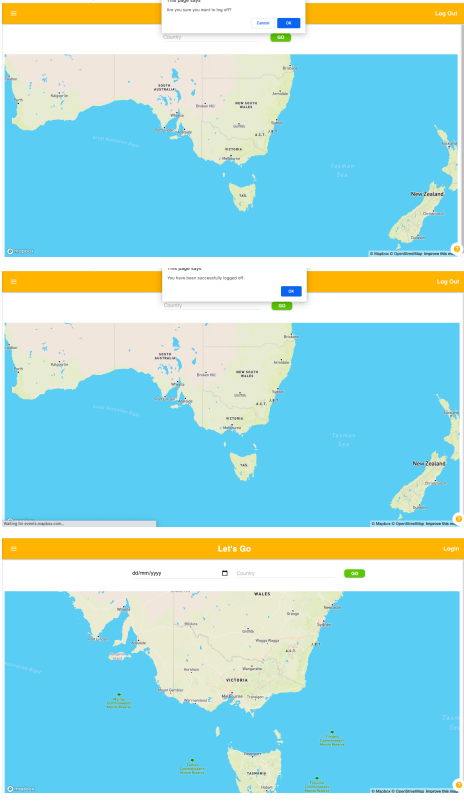
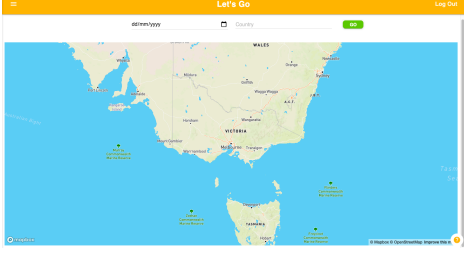
1. The user should be able to search for valid countries.
2. A dropdown list of countries of all of the valid countries should be provided as the user enters a country's name.
3. The user should be able to select only one country.
4. The user should be informed if the country inputted is invalid.
5. The user should be panned to the user-selected country.
6. The user should be able to drag and zoom in on the map.
7. All the airports within the country should be displayed.
8. Each airport should be marked with a marker symbol to indicate that they are an airport.
9. The airports should be labelled with the correct name when clicked on.
 - a. The user should be able to close the label.
10. All the routes should be domestic/within the country.
11. Only the airports that have routes should be shown.
12. All routes should be indicated as continuous red lines.
13. When the user enters a new country, the routes and airports in the previous country should disappear and the user should be taken to the new country with the routes and airports in that country shown.

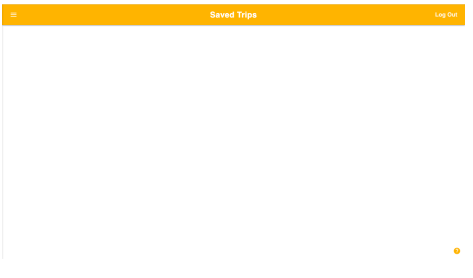
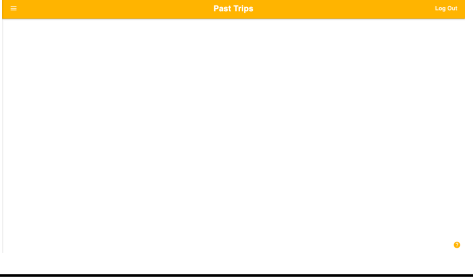
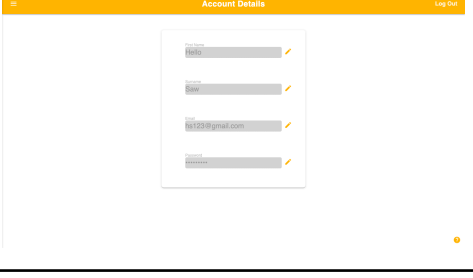
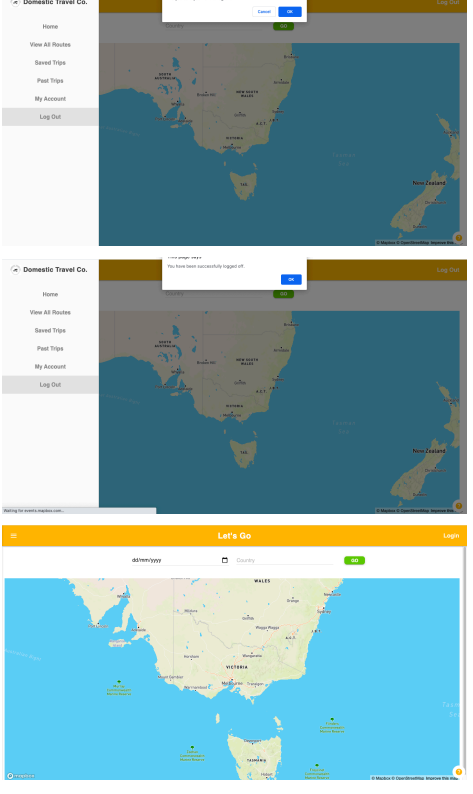
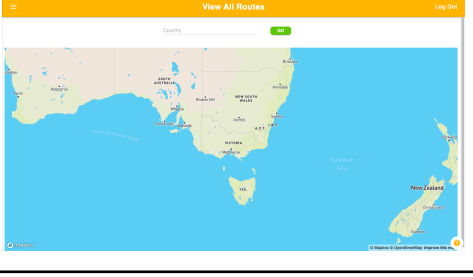
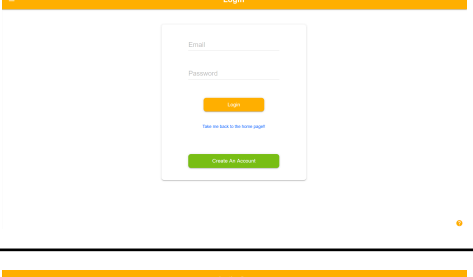
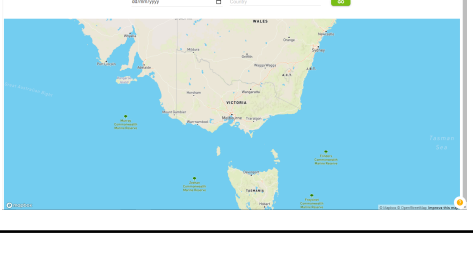
2.0 System / Integration Tests

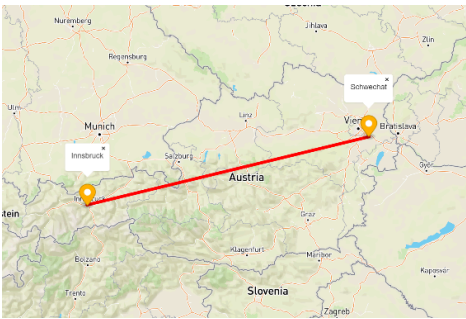
Test Number	Reason for Test	Input	Expect Output	Actual Output	Test Passed?
1	To ensure that all the markers/airports are within the country.	"Austria"	Expect 6 airports to be shown with a orange marker and they are all within the country		Passed
2	To ensure that all the routes are within the country.	"Austria"	Expect that there are six routes among the airports within the country		Passed
3	To ensure that all the map is panned to the inputted country.	"Austria"	The user is panned to Austria		Passed

4	To ensure that all the markers/airports have at least one route connected to it.	“Austria”	All the markers/airports have at least one route linked to it		Passed
5	To ensure that all popups with the airport name appear when the marker/airport is clicked on.	“Austria”. Click on the far right marker.	A popup pointing to the marker with the airport’s name (Schwechat) and a close button in the top right corner that closes the popup when pressed.		Passed

3.0 Unit Tests

Test Number	Reason for Test	Input	Expected Output	Actual Output	Test Passed?
1	To ensure that the navigation bar is appropriate - depending on whether the user is logged in or not	User is logged in	The navigation bar appears on the right of the screen. The logo and company name on the first line. ‘Home’, ‘View all Routes’,'Saved Trips’, ‘Past Trips’, ‘My Account’, ‘Log out’ buttons shown		Passed
2		User is logged out	The logo and company name on the first line. ‘Home’, ‘View all Routes’, ‘Login’ and ‘Create Account’ buttons shown		Passed
3	To ensure that all the navigation bar and drawer links when the user is logged in are working	Press the ‘logout’ button in the header	The user will be prompt to confirm logout and after confirmation, a informative popup will appear and the user will be redirected to the home page		Passed
4		Press ‘Home’ in the navigation draw	The user will be redirected to the home page		Passed

5		Press ‘Saved Trips’ in the navigation draw	The user will be redirected to the save trips page		Passed
6		Press ‘Past Trips’ in the navigation draw	The user will be redirected to the past trips page		Passed
7		Press ‘My Account’ in the navigation draw	The user will be redirected to the account details page		Passed
8		Press ‘Log out’ in the navigation draw	The user will be prompt to confirm logout and after confirmation, a informative popup will appear and the user will be redirected to the home page		Passed
9		Press the ‘View all Routes’ in the navigation draw	The view all routes page will refresh, removing all current inputs and outputs		Passed
10	To ensure that all the navigation bar and drawer links when the user is logged out/a guest users are working	Press the ‘login’ button in the header	The user is directed to the login page		Passed
11		Press ‘Home’ in navigation drawer	The user is directed to the home page		Passed

23	To ensure that the map can be resetted	Load the page, view all the routes in “New Zealand” and then in console, call ‘resetTheMap();’	resetTheMap function is called, when viewing all routes so that map can be resetted	<pre>> resetTheMap < f resetTheMap() { map = new mapboxgl.Map({ container: 'map', center: [144.9648731, -37.8182711], zoom: 4, style: 'mapbox://styles/mapbox/streets-v9' }); }</pre>	Passed
24	To ensure that the function storingAirportsInformation can successfully store the airports information	Define: airportsData (see below [1]) Call: storingAirportsInformation and pass it. Check the value of dataOfAirports.	All the relevant information of the airports will be stored in dataOfAirports.	<pre>_airline: "OS" ▼ _airportIds: Array(2) 0: "INN" 1: "VIE" length: 2 ► __proto__: Array(0) ▼ _cities: Array(2) 0: "Innsbruck" 1: "Vienna" length: 2 ► __proto__: Array(0) ▼ _latitude: Array(2) 0: "47.260219" 1: "48.110278" length: 2 ► __proto__: Array(0) ▼ _longitude: Array(2) 0: "11.343964" 1: "16.569722" length: 2 ► __proto__: Array(0) _numOfTransits: 0 ▼ _routeAirports: Array(2) 0: "Innsbruck" 1: "Schwechat"</pre>	Passed
25	To ensure that the function generateAirportsAndRoutes can successfully generate the airports and routes given	Define: allRoutesData (see below [2]) Call: generateAirportsAndRoutes and pass it. Check the appropriate airports, routes and airport popups have been shown.	It should generate a marker for both Innsbruck and Schwechat airport, have a popup for these markers that are closeable and a route line between them.		Passed

[1]

airportsData = [{IATA-FAA: "INN", airportId: "1610", altitude: "1906", city: "Innsbruck", country: "Austria", latitude: "47.260219", longitude: "11.343964", name: "Innsbruck"}, {IATA-FAA: "VIE", airportId: "1613", altitude: "600", city: "Vienna", country: "Austria", latitude: "48.110278", longitude: "16.569722", name: "Schwechat"}].

[2]

allRoutesData = [{airline: "OS", airlineId: 491, codeshare: "Y", destinationAirport: "VIE", destinationAirportId: 1613, equipment: "F70 100 DH4 320", sourceAirport: "INN", sourceAirportId: 1610, stops: 0}].

4.0 Test Result Summary

In total, five integration tests and twenty five unit tests were conducted to address any issues with regards to the feature “View all routes” in a country. The integration tests conducted checked for proper functionality of the page, for instance mapping all the markers, routes within the country that is selected. The unit tests were conducted as in to check the working of small portions of codes, each unit test tested a function that is relevant to the “view all routes” page. For instance to verify the valid working of the search button in the page, a number of false/dummy values were entered, as a result of which a pop was shown that input was invalid, validating that the search button has passed the test. Another instance for doing the unit tests was to check and valid whether the navigation bar would change depending upon whether the user is either logged in or not. It was also tested whether once a user leaves this page, if the map could be resetted by using the ‘resetThemap()’ function. Finally it was also checked whether the details of the airports were being stored appropriately in the local storage. Out of all the tests that were performed, none of the tests failed and all the outputs of these tests matched the required output.